

Stars

🏷 Tags:

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Codes

AI Code Tips

Description

On a clear moon-less night, you can see millions of stars glimmering in the sky. Faced with this overwhelming number, the Greeks started nearly 2,000 years ago to bring some order to the chaos. They identified groups of stars, called constellations, and gave them names, mostly from the Greek mythology, that are still in use today. Examples are "Ursa Minor", "Pisces", "Cancer", and many others.

Given a sketch of the constellation, it is not easy for the amateur to actually find the constellation in the sky. Moreover, simple constellations, such as "Triangulum" (triangle,) which consists of only three stars, may appear several times in the sky. Again, singling out the "correct" occurrence is not easy.

Traditionally, maps were printed for just this purpose. But in this problem, we will see how the computer can help us find constellations in the sky.

You will be given a star map; for simplicity this will be a collection of points in the plane, each having a certain brightness associated with it. Then, given a constellation, also as a set of points in the plane, you are to determine:

- the number of occurrences of the constellation in the star map, and
- the position of the brightest occurrence, if one exists. (The rationale behind this is as follows: if a constellation seems to appear several times in the sky, the brightest one is most likely to be the real one, since it is the most eye-catching one.)

An occurrence is a subset of stars from the map that forms a (possibly) arbitrarily rotated and/or scaled copy of the stars in the constellation.

The brightness of an occurrence is the average brightness of the stars it consists of, i.e. the sum of individual brightnesses divided by the number of stars in the constellation.

Input

On a clear moon-less night, you can see millions of stars glimmering in the sky. Faced with this overwhelming number, the Greeks started nearly 2,000 years ago to bring some order to the chaos. They identified groups of stars, called constellations, and gave them names, mostly from the Greek mythology, that are still in use today. Examples are "Ursa Minor", "Pisces", "Cancer", and many others.

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Output

For each star map first output the number of the map (``Map #1 '`, ``Map #2 '`, etc.) on a line of its own.

For each constellation, in the same order as in the input, output first its name and how many times it occurs in the map on one line, as shown in the output sample.

If there is at least one occurrence, output the position of the brightest occurrence by listing the positions of the stars that form the brightest occurrence. The star positions have to be printed in ascending *x*-order. Positions having the same *x*-coordinates must be sorted in ascending *y*-order. If there are several equally bright solutions, output only one of them. Adhere to the format shown in the sample output.

Output a blank line before each constellation and a line of 5 dashes (``----- '`) after every star map.

Samples

input	Copy
<pre>6 1 2 1 2 1 4 2 4 3 3 2 1 4 1 5 4 3 2 2 3 Triangulum 1 1 3 1 2 4 4 Cancer 1 3 4 3 6 1 7 5 0</pre>	
output	Copy
<pre>Map #1 Triangulum occurs 2 time(s) in the map. Brightest occurrence: (1,2) (4,1) (4,3) Cancer occurs 0 time(s) in the map. -----</pre>	

Submit

Codes

[HZNUOJ](#) is based on [HUSTOJ](#)

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